

Use cases

Here are some common use cases of open source AI in IoT.

Smart cities: AI-powered IoT devices can manage traffic flow, reduce energy consumption, and improve public safety through predictive analytics and real-time data processing.

Supply chain optimisation: AI can analyse data from IoT sensors to improve inventory management, predict delivery delays, and enhance logistics operations.

Autonomous vehicles: Open source AI helps IoT sensors in autonomous vehicles process real-time data for navigation, collision avoidance, and route optimisation.

Healthcare monitoring: Wearable IoT devices integrated with AI can monitor patient vitals, track health trends, and alert healthcare providers about potential issues.

Now is the perfect time for businesses and developers to embrace the power of open source AI frameworks in

their IoT projects. By exploring various open source AI frameworks, they can unlock new levels of efficiency, innovation, and scalability. These frameworks make advanced AI accessible even for resource-constrained devices, enabling smarter and more autonomous systems.

Don't miss out on the opportunity to stay ahead of the curve—explore these open source AI integrations today and lead the charge in the next wave of IoT innovation! **END** 🐧

 **By: Dhaval Gajjar**

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Low-code platforms like Node-RED are changing how IoT applications are created. Users can design systems by dragging and dropping pre-built blocks, each representing a function like data collection or communication with devices. For example, a student can set up a temperature monitoring system by linking blocks for sensors, data logging, and alerts—without writing any code. This simplicity makes IoT development accessible to people with basic computer knowledge. It's especially helpful for students, small businesses, and hobbyists who want to explore IoT without deep programming skills.

Low-code platforms also make it easier to develop practical IoT systems. For instance, in smart agriculture, students can design systems to monitor soil moisture and control irrigation automatically, helping farmers save water. In smart homes, hobbyists can build systems to control appliances like lights or security cameras. In cities like Delhi, where air quality is a concern, these platforms can help create IoT systems to track pollution levels in real-time.

By simplifying the development process, these platforms are democratising IoT and empowering the next generation of pioneers eager to explore new technologies.

Challenges and the future

Open source IoT has many advantages, but it also faces some significant challenges. One of the biggest concerns is security. Since open source software is publicly available, it can be vulnerable if not regularly updated and properly managed. Developers need to ensure that security measures are strong to protect both the devices and the data they handle. Another challenge is interoperability. Devices from different manufacturers may not always work well together, which can limit their usefulness. Creating standardised communication protocols is key to overcoming this issue.

Despite these challenges, the future of open source IoT looks promising. It's opening up opportunities for developers, small

businesses, and startups by cutting down on costs and promoting collaboration. Hackathons, community-driven projects, and innovation programs are helping to push forward new solutions and ideas.

As more people contribute to open source projects, we can expect faster innovation, better security, and lower costs. Open source IoT is also likely to play a big role in meeting sustainability goals. By providing more efficient, eco-friendly solutions, it can help reduce waste and energy consumption.

Looking ahead, open source IoT will continue to drive change. It's a key part of technological growth across industries and will help improve how we live. Emerging technologies like edge computing, artificial intelligence, blockchain security, and digital twins are changing how devices communicate and perform tasks. While challenges like security risks and device compatibility still exist, the potential of open source IoT to address them is clear. **END** 🐧

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